

[ACJC/2025/Q4]

4 Name your Jupyter Notebook as:

TASK4_<your name>_<centre number>_<index number>.ipynb

A community centre keeps records about its courses. The community centre wants to create a suitable database to store the data and allow them to run searches for specific data. The database will have four tables: a table to store data about the instructors, a table about the customers, a table about the courses, and a table to show which customer has registered for which course.. The fields in each table are:

Instructor:

- InstructorID – instructor's unique ID
- Name – the instructor's name

Customer:

- CustomerID – the customer's unique ID
- Name – the customer's name

Course:

- CourseID – unique course ID, for example, D123
- Name – the name of the course
- InstructorID – the ID of the instructor conducting the course
- StartDate – the starting date of the course in YYYYMMDD format
- Price – the price of the course in dollars

Registration:

- CustomerID – the customer's ID
- CourseID – the course ID
- Paid – an integer value, where 0 means the customer has not paid and 1 means the customer has paid

For each of the sub-tasks 4.1 and 4.2, add a comment statement at the beginning of the code, using the hash symbol '#' to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 4.1
        Program code
        Output:
```

Task 4.1

Write a Python program that uses SQL code to create the database COURSES with the four tables given. Define the primary and foreign keys for each table. [6]

Task 4.2

The text files COURSE.txt, INSTRUCTOR.txt, CUSTOMER.txt and REGISTRATION.txt contains the comma-separated values for each of the tables in the database.

Write a Python program to read the data from each file and then store each item of data in the correct place in the database. [5]

[ACJC/2025/Q4 Cont']

Task 4.3

Write a Python program and the necessary files to create a web application that asks the user for a customer name, and displays the following data for all courses that the customer has registered for.

- Course name
- Instructor name
- Whether the customer has paid for the course

The program should return an HTML document that enables the web browser to display a table with the required data.

Save your program code as

TASK4_3_<your name>_<centre number>_<index number>.py

with any additional files/subfolders as needed in a folder named

TASK4_3_<your name>_<centre number>_<index number>

[11]

Run the web application, entering "Tan Ah Seng" as the customer name.

Save the webpage output as:

TASK4_3_<your name>_<centre number>_<index number>.html

[2]